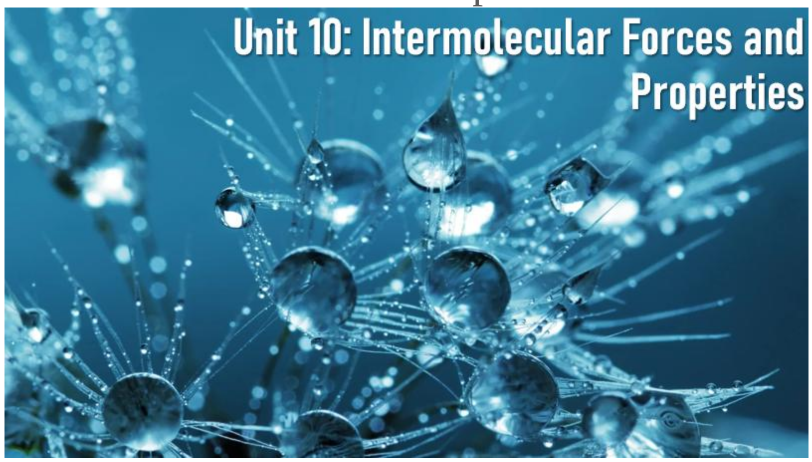
Name: ____ Period: ____________

Unit 10: Intermolecular Forces and Properties

AP Classroom Topic 3, Partial 7

Unit Map



Learning Targets & Success Criteria

Textbook Reference: Zumdahl, 5, 10, 11, 16

Date	Day	Learning Target	Activities	Work on this
3/25-26	U10D1	How do gases behave?	Wrap up bonding practice Begin U10 IMF notes	REVIEW FOR AP EXAM
3/27-30	U10D2	What are intermolecular forces?		U10 Textbook Probs #1 REVIEW FOR AP EXAM
3/31-4/1	U10D3	How do IMFs help predict properties?	IMF Card Sort	U10 Textbook Probs #1 REVIEW FOR AP EXAM
4/2-6	U10D4	How do IMFs determine phase and solubility?	Lab: Calming Bottle	U10 Textbook Probs #1 REVIEW FOR AP EXAM
4/7-8	U10D5	What determines precipitation?	~IMFs Quiz	U10 Textbook Probs #2 REVIEW FOR AP EXAM
4/9-10	U10D6	How does solubility relate to equilibrium? What is the common ion effect?		U10 Textbook Probs #2 REVIEW FOR AP EXAM
4/13-14	U10D7	What have we learned about aqueous equilibria?	Ksp Quiz	REVIEW FOR AP EXAM

	Unit 10 Intermolecular Forces and Solutions (AP Chem Topic 3, 7)
3.1	**Intermolecular and Interparticulate Forces**
3.1A	Explain the relationship between the chemical structures of molecules and the relative strength of their intermolecular forces when: i. The molecules are of the same chemical species. ii. The molecules are of two different chemical species.
	3.1.A.1: London dispersion forces are a result of the Coulombic interactions between temporary, fluctuating dipoles. London dispersion forces are often the strongest net intermolecular force between large molecules.
	i. Dispersion forces increase with increasing contact area between molecules and with increasing polarizability of the molecules. ii. The polarizability of a molecule increases with an increasing number of electrons in the molecule and the size of the electron cloud. It is enhanced by the presence of pi bonding. iii. The term "London dispersion forces" should not be used synonymously with the term "van der Waals forces."
	3.1.A.2: The dipole moment of a polar molecule leads to additional interactions with other chemical species.
	Dipole-induced dipole interactions are present between a polar and nonpolar molecule. These forces are always attractive. The strength of these forces increases with the magnitude of the dipole of the polar molecule and with the polarizability of the nonpolar molecule.
	ii. Dipole-dipole interactions are present between polar molecules. The interaction strength depends on the magnitudes of the dipoles and their relative orientation. Interactions between polar molecules are typically greater than those between nonpolar molecules of comparable size because these interactions act in addition to London dispersion forces.
	iii. Ion-dipole forces of attraction are present between ions and polar molecules. These tend to be stronger than dipole-dipole forces.
	3.1.A.3: The relative strength and orientation dependence of dipole-dipole and ion-dipole forces can be understood qualitatively by considering the sign of the partial charges responsible for the molecular dipole moment, and how these partial charges interact with an ion or with an adjacent dipole.
	3.1.A.4: Hydrogen bonding is a strong type of intermolecular interaction that exists when hydrogen atoms covalently bonded to the highly electronegative atoms (N, O, and F) are attracted to the negative end of a dipole formed by the electronegative atom (N, O, and F) in a different molecule, or a different part of the same molecule.
	3.1.A.5: In large biomolecules, noncovalent interactions may occur between different molecule or between different regions of the same large biomolecule.
3.2	**Properties of Solids**
3.2A	Explain the relationship among the macroscopic properties of a substance, the particulate-level structure of the substance, and the interactions between these particles.
	3.2.A.1: Many properties of liquids and solids are determined by the strengths and types of intermolecular forces present. Because intermolecular interactions are overcome completely when a substance vaporizes, the vapor pressure and boiling point are directly related to the strength of those interactions. Melting points also tend to correlate with interaction strength, but because the interactions are only rearranged, in melting, the relations can be more subtle.
	3.2.A.2: Particulate-level representations, showing multiple interacting chemical species, are a useful means to communicate or understand how intermolecular interactions help to establish macroscopic properties.
	3.2.A.3: Due to strong interactions between ions, ionic solids tend to have low vapor pressures, high melting points, and high boiling points. They tend to be brittle due to the repulsion of like charges caused when one layer slides across another layer. They conduct electricity only when the ions are mobile, as when the ionic solid is melted (i.e., in a molten state) or dissolved in water or another solvent.
	3.2.A.4: In covalent network solids, the atoms are covalently bonded together into a three-dimensional network (e.g., diamond) or layers of two-dimensional networks (e.g., graphite). These are only formed from nonmetals and metalloids: elemental (e.g., diamond, graphite) or binary compounds (e.g., silicon dioxide and silicon carbide). Due to the strong covalent interactions, covalent solids have high melting points. Three-dimensional network solids are also rigid and hard, because the covalent bond angles are fixed. However, graphite is soft because adjacent layers can slide past each other relatively easily.
	3.2.A.5: Molecular solids are composed of distinct, individual units of covalently-bonded molecules attracted to each other through relatively weak intermolecular forces. Molecular solids generally have a low melting point because of the relatively weak intermolecular forces present between the molecules. They do not conduct electricity because their valence electrons are tightly held within the covalent bonds and the lone pairs of each constituent molecule. Molecular solids are sometimes composed of very large molecules or polymers.
	3.2.A.6: Metallic solids are good conductors of electricity and heat, due to the presence of free valence electrons. They also tend to be malleable and ductile, due to the ease with which the metal cores can rearrange their structure. In an interstitial alloy, interstitial atoms tend to make the lattice more rigid, decreasing malleability and ductility. Alloys typically retain a sea of mobile electrons and so remain conducting.
	3.2.A.7: In large biomolecules or polymers, noncovalent interactions may occur between different molecules or between different regions of the same large biomolecule. The functionality and properties of such molecules depend strongly on the shape of the molecule, which is largely dictated by noncovalent interactions.
3.10	**Solubility**
3.10A	Explain the relationship between the solubility of ionic and molecular compounds in aqueous and nonaqueous solvents, and the intermolecular interactions between particles.
	3.10.A.1: Substances with similar intermolecular interactions tend to be miscible or soluble in one another.
7.11	**Introduction to Solubility Equilibria**
7.11A	Calculate the solubility of a salt based on the value of Ksp for the salt.
	7.11.A.1: The dissolution of a salt is a reversible process whose extent can be described by Ksp, the solubility-product constant.
	7.11.A.2: The solubility of a substance can be calculated from the Ksp for the dissolution process. This relationship can also be used to predict the relative solubility of different substances.
	7.11.A.3: The solubility rules (see 4.7.A.5) can be quantitatively related to Ksp, in which Ksp values >1 correspond to soluble salts.
	7.11.A.4: The molar solubility of one or more species in a saturated solution can be used to calculate the Ksp of a substance.
7.12	**Common Ion Effect**
7.12A	Identify the solubility of a salt, and / or the value of Ksp for the salt, based on the concentration of a common ion already present in solution.
	7.12.A.1: The solubility of a salt is reduced when it is dissolved into a solution that already contains one of the ions present in the salt. The impact of this "common-ion effect" on solubility can be understood qualitatively using Le Châtelier's principle or calculated from the Ksp for the dissolution process.
8.11	**pH and Solubility**
8.11A	Identify the qualitative effect of changes in pH on the solubility of a salt.
	8.11.A.1: The solubility of a salt is pH sensitive when one of the constituent ions is a weak acid, a weak base, or the hydroxide ion. These effects can be understood qualitatively using Le Châtelier's principle.

Ideal Gases: Not Attractive at all

Kinetic Molecular Theory of Gases

- a simple model that attempts to explain properties of an _____________ gas
- based on an assumption about the particles: that they experience NO ____________________________ ____________________________ at all

Ideal gases consist of particles that have the following properties:

1. The particles are so small compared to the distances between them that the _______________ of the individual particles can be assumed to be ______________________ (zero), but their _____________________ is NOT _____________________.

2. The particles are in ______________________ ____________________. The ________________ of the _______________ of the particles with the ____________ of the container are the cause of the ________________ exerted by the gas.

3. The particles are assumed to exert ______ _______________________ on each other; they are assumed neither to _____________________ nor to _______________ each other.

4. The _____________________ ________________________ ________________________ _________________ of a collection of gas particles is assumed to be directly proportional to the ____________ ______________ of the gas.

-Because ________________ gases conform to these assumptions, we can describe its P, V, n, and T with simple mathematical models, the Ideal Gas Law: **PV = nRT**

-BUT __________ gases don't conform to these assumptions!!!

Application: Molar Mass from Gases

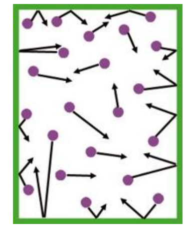
- Recall that the Ideal Gas Law, PV = nRT, is useful for determining _________________ of a gas for stoichiometry and other uses
- Also, when paired with mass data, the __________ __________ of a gas can be determined:

MM = mass/moles = m/n

Practice: A sample of a certain gas occupies 8.5 L at 1 ^{o} C and 761 Torr and has a mass of 6.1 grams. What is the molar mass of the gas?

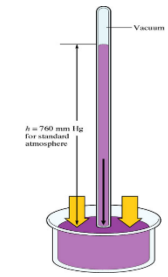
Mystery Gas! In the basement of the old chemistry building at Mystery University, a gas cylinder has been found. The gas needs to be disposed of properly, but in order to do that, the identity of the gas needs to be determined. The sealed 7.5L gas cylinder is held at 299K. The pressure gauge reads 26500 Torr. The mass in the cylinder is 404 g and through other chemical tests, it is known to be a halogen, what halogen is it?

Understanding Gas Pressure



This is a schematic diagram showing gas molecules in a container. The molecules are constantly moving in random directions. When a molecule hits the container wall (green), it exerts a tiny _____________ on the wall. The sum of these tiny forces, divided by the interior surface area of the container, is the _______________. (Longer arrows indicate ____________ velocities, shorter arrows indicate ______________ velocities)

Why aren't all the molecules moving with the same velocity and how is this related to temperature?



If the temperature increases, what happens to the velocities? To the pressure?

Barometer- measures ____________________ in ________________

MCQ Practice:

1. Two **flexible** containers for gases are at the same temperature and pressure. One holds 0.50 gram of hydrogen and the other holds 8.0 grams of oxygen. Which of the following statements regarding these gas samples is **FALSE**?
a. The volume of the H_2 container is the same as the volume of the O_2 container.
b. The number of molecules in the H_2 container is the same as the number of molecules in the O_2 container.
c. The average kinetic energy of the H_2 molecules is the same as the average kinetic energy of the O_2 molecules.
d. The average speed of the H_2 molecules is the same as the average speed of the O_2 molecules.

2. A sample of an ideal gas is cooled from 50.0 ^{0}C to 25. 0 ^{0}C in a sealed container of constant volume. Which of the following values for the gas will decrease?
a. The average molecular mass of the gas
b. The average distance between the molecules
c. The average speed of the molecules
d. The average distance between the molecules and the average speed of the molecules

Molar Volume
Molar Volume: At standard temperature and pressure, ____ mol of an ideal gas = ________ L

STP = _______ ^{o}C and __________ atm

DANGER! DANGER! DANGER! In AP Chem FRQs, gases are almost NEVER at STP, so you should almost NEVER use this 22.4! It is more likely in MCQs.

Real (nonideal) Gases

- Ideal gas behavior assumes no attractions or repulsions between particles. No gas exactly follows the ideal gas law.
- The ___________ the IMFs (attractions) in a substance, the _____________ ideal a gas behaves. ________________, _______________________ particles are the most ideal gases.
- Also, real gas exhibits behavior closest to ideal behavior at ___________ _______________ (widely spaced particles) and ___________ ____________________ (highly energetic molecules). Students also behave most ideally under these conditions. (Summer vacation!)

- If the pressure is too _______________, the particles are not separated by an infinite distance and can interact, slowing/changing their trajectory.

- If the temperature is too _______________, the particles slow their speed and can interact, slowing/changing their trajectory.

Practice: Which of the following gases deviates most from ideal behavior?

a. SO_2 c. N_2
b. CH_4 d. H_2

Effusion and Diffusion

Diffusion- _____________________ of gases- happens because gases move randomly
Effusion- the passage of a gas through a tiny orifice into an evacuated chamber- happens because of differences in __________________

Rates are determined by ________________ - lighter gases effuse & diffuse _______________ than heavier gases.
This affects _____________________ of gases

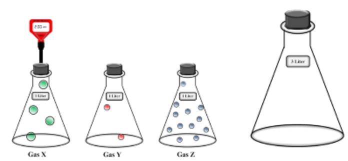
Practice: Equal number of moles of He(g), Ar(g), and Ne(g) are placed in a glass vessel at room temperature. If the vessel has a pinhole-sized leak, which of the following will be true regarding the relative values of the partial pressures of the gases remaining in the vessel after some of the gas mixture has effused?

a. $P_{He} < P_{Ne} < P_{Ar}$ c. $P_{Ne} < P_{Ar} < P_{He}$
b. $P_{He} < P_{Ar} < P_{Ne}$ d. $P_{Ar} < P_{He} < P_{Ne}$

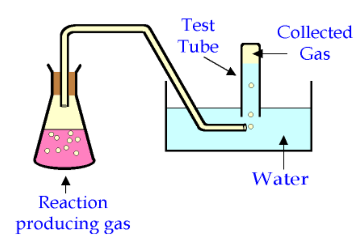
Dalton's Law of Partial Preassures

For a ___________________ of gases in a container, the ________________ pressure exerted is the ____________ of the pressures that each gas would exert if it were alone.

$$P_{tot} = P_1 + P_2 + P_3 + \ldots$$



Practice: Each one liter flask pictured above is sealed and contains a different noble gas at 298 K. The pressure of Gas *X* in the first flask is 2.0 atm. The ratio of the numbers of *X*, *Y* and Z atoms in the flasks is 2:1:6, respectively. A student then combines all the gases into a previously evacuated 3.0 L flask. What is the total pressure of the gases at 298 K? Hint- read this problem carefully. A. 3.0 atm B. 4.5 atm C. 9.0 atm D. 18 atm



Application: Gases Collected over Water

When gases are collected over water, we must adjust for the pressure of the water vapor.

$$P_{H2O} + P_{gas} = P_{total}$$

EXAMPLE 5.11 Collecting Gases over Water

In order to determine the rate of photosynthesis, the oxygen gas emitted by an aquatic plant is collected over water at a temperature of 293 K and a total pressure of 755.2 mmHg. Over a specific time period, a total of 1.02 L of gas is collected. What mass of oxygen gas (in grams) is formed?

Intro to IMFs: Sooooo Attractive!

<u>Intra</u>molecular forces: the attraction that results when electrons are given, taken, or shared to form _____ bond.

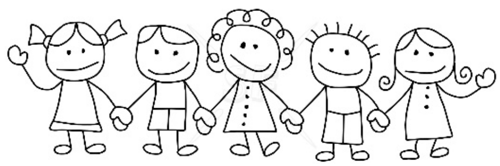
- Ionic bonds
- Covalent bonds
- Metallic bonds

<u>Inter</u>molecular forces (IMFs): the attraction ____________________ two or more distinct molecules or particles.

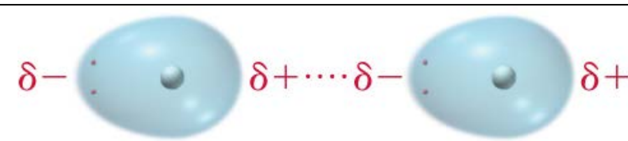
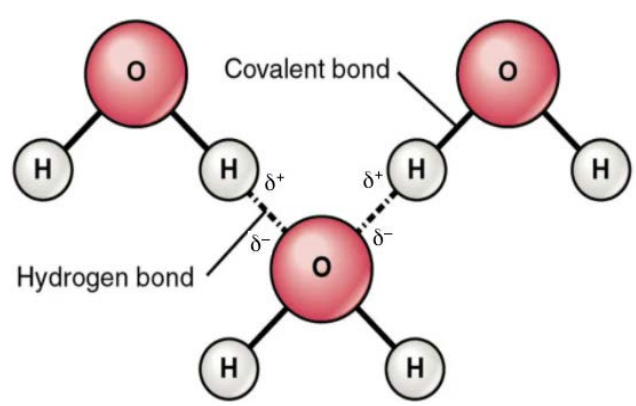
- In a physical change, only _________________________ forces are broken because no chemical bonds are broken (same substance).
- In a chemical change, _________________________ forces are broken as chemical bonds are broken/formed (new substance).

<u>Intra</u>molecular forces are much, MUCH stronger than <u>inter</u>molecular forces!

Think of the children!

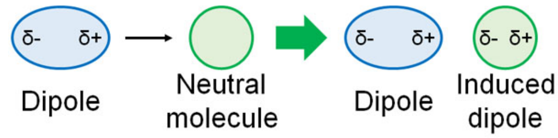
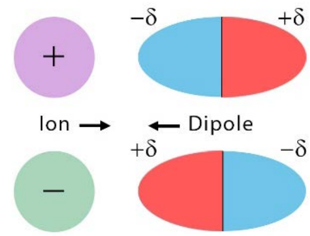


Types of IMFs

Type	Present in	Molecular perspective
London Dispersion Forces/ Van der Waals Forces (LDFs)	• ________ molecules and atoms (nonpolar and polar) • Strength depends on total _____ of electrons in an atom/molecule	 instantaneous dipoles
Dipole-dipole attractive forces	• Between ______________ molecules (which already have a dipole moment)	
Hydrogen "bonding" attractive forces	• Between H bonded to F, O, or N on one molecule and **<u>lone pair</u>** <u>on N, O, or F of another molecule</u> → they are electronegative e_________ ! → Hydrogen "bonds" are ______ actual bonds (intramolecular forces); the name is very misleading!	

If comparing **similar-sized** particles: **Hydrogen bonding IMFs > dipole-dipole IMFs > LDFs**

Mixed IMFs! Attractive Forces Between Two __________________ Compounds

Type	Present in	Molecular perspective
Dipole – induced dipole attractive forces	**<u>Polar</u>** <u>molecules</u> combined with **<u>Nonpolar</u>** <u>molecules</u>	
Ion – dipole attractive forces	**<u>Ions/ Ionic</u>** <u>Compounds</u> combined with Polar molecules	

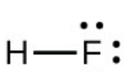
If comparing **similar-sized** particles:

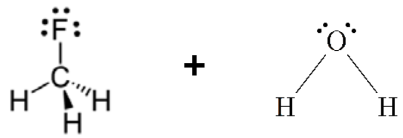
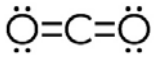
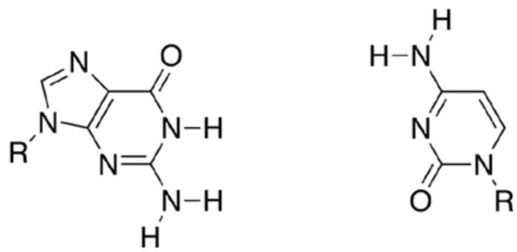
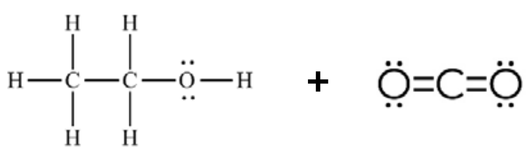
Ion – Dipole **>** Hydrogen bonding **>** Dipole – dipole **>** Dipole – Induced Dipole **>** LDFs

In Summary

	Non-polar, single elements	*polar + non-polar*	polar no H – FON	polar with H – NOF	*polar + ions*
London dispersion forces (LDFs or Van der Waals)					
Dipole – Induced Dipole (multiple compounds)					
Dipole – Dipole					
Hydrogen “bonding” attractive forces					
Ion – Dipole (multiple compounds)					

Let’s Practice!

Molecule 1. Label each as polar, non-polar, or an ion 2. If polar, is it NOF?	**IMFs present?** 1. Which IMFs are present between the compounds? 2. Underline the dominant (strongest) IMF exhibited
$CH_3CH_2CH_3$	
	

	
	
 Guanine **+** Cytosine	
	

Multiple Choice Practice

1. The forces of attraction between the permanent partially positive regions of one molecule and permanent partially negative regions of another nearby molecule are called ____________.

   a. covalent bonds
   b. hydrogen bonding
   c. London dispersion forces
   d. dipole−dipole forces

2. A polar molecule is one in which:

   a. ions exist
   b. only london dispersion forces exist
   c. it has a VSEPR structure that is symmetrical
   d. electron density is unequally distributed

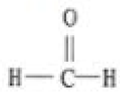
3. Which of the following processes involves breaking intermolecular forces?

   a. $H_2(g) + Cl_2(g) \rightarrow 2\ HCl(g)$
   b. $2\ C_2H_6(g) + 7\ O_2(g) \rightarrow 4\ CO_2(g) + 6\ H_2O(g)$
   c. $I_2(g) \rightarrow 2\ I(g)$
   d. $H_2O(s) \rightarrow H_2O(l)$

4. What is the dominant intermolecular force in CH_3OH?

   a. London dispersion forces
   b. Ion-dipole attraction
   d. Dipole-dipole attraction
   e. Hydrogen bonding

5. Which one of the following substances will have hydrogen bonding as one of its intermolecular forces?

(A)

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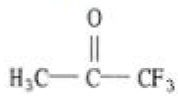
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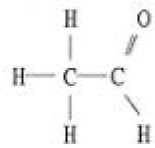
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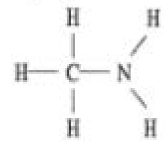
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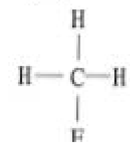
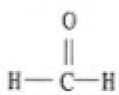


6. Circle ALL of the following substances which <u>cannot</u> hydrogen bond with another molecule of the **same** substance, but would be capable of hydrogen bonding with a **different** molecule that has H directly bonded to fluorine, oxygen, or hydrogen?

(A)

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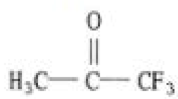
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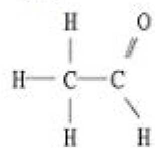
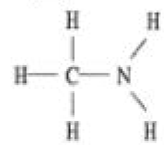

(D)

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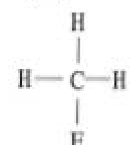


The Language of IMFs (i.e. How to FR)

Notes about Language: Talking about IMFs can be tricky! And of course, IMF-based questions are VERY common on free response portion of the AP test, so it's essential to carefully choose your language to avoid losing points.

Most IMF points are lost when student inadvertently imply (or, worse, directly state) that during state changes:

1. ________________ bonds are being broken during state changes **(NOPE!)**
2. Molecules are being ____________ apart **(NOPE!)**

How to Talk about IMFS	How to Talk about Bonds
• ______________________ IMFs o Avoid breaking IMFs • ______________________ molecules o NEVER break apart molecules • IMFs are ______________ molecules	• ______________ bonds • Bonds are _____________molecules

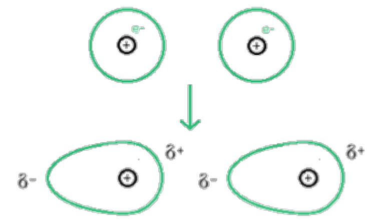
Free Response Practice: What <u>**type**</u> of intermolecular OR intramolecular attractive forces must be overcome for each of the following processes to occur?

1. $CO_2(g) \rightarrow C(s) + O_2(g)$, $\Delta H = 393.5$ kJ/mol

2. $CO_2(s) \rightarrow CO_2(g)$, $\Delta H = 25.2$ kJ/mol

<u>A Closer Look at London Dispersion Forces (LDFs):</u> Induced Dipole – Induced Dipole Attraction

LDFs are determined by the ______________________ of a molecule (i.e. how much the electron cloud can temporarily be shifted)

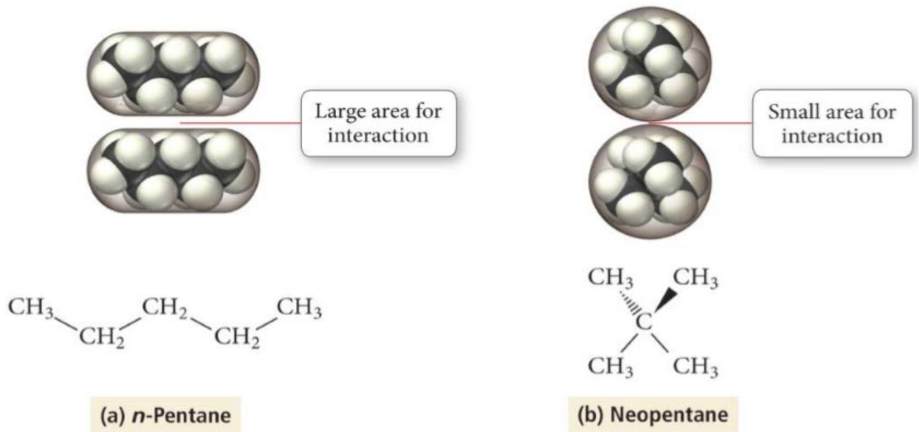


➔ an electron cloud, even in a nonpolar molecule, can ____________________ shift, causing one side of the molecule to be more negative than another

➔ this temporary dipole can ___________ (cause) a temporary dipole on a neighboring molecule (hence the name "induced dipole")

What can increase LDFs?

1. <u>Greater electron cloud</u> (more electrons): molecule is ________ polarizable = ____ LDFs
2. <u>Increase in molar mass</u> (implies more electrons): molecule is ________ polarizable = ____ LDFs
3. <u>Increase in surface-to-surface contact area</u>: ____ induced dipole = ____ LDFs



Boiling point: 36.1°C

Boiling point: 10.0°C

Be careful!

- When non-polar substances with ________ London dispersion forces have a considerably ________________ (and thus very polarizable) electron cloud than the polar molecules, the LDFs can be quite substantial and be **STRONGER** than hydrogen bonding forces or dipole-dipole forces (!!)

Example: Cl_2 has a higher boiling point that HCl. Explain.

Let's Practice!

1. Rank the following in order of increasing LDFs: CH_3CH_3, CH_4, $CH_3CH_2CH_3$

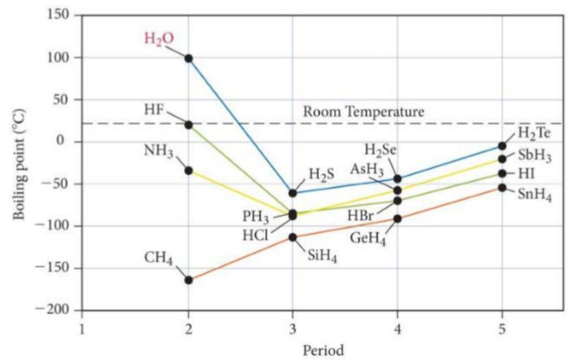
2. Rank the following in order of increasing LDFs: Br_2, F_2, Cl_2, I_2

A Closer Look at Hydrogen Bonding Attractive Forces

Note: Hydrogen "bonds" are ______actual bonds (intramolecular forces), and thus the name is very misleading!

Hydrogen "Bonding": force of attraction between hydrogen atom bonded to a small highly electronegative atom (N, O, and F) and the unshared electron pair on another electronegative atom of N, O, or F.

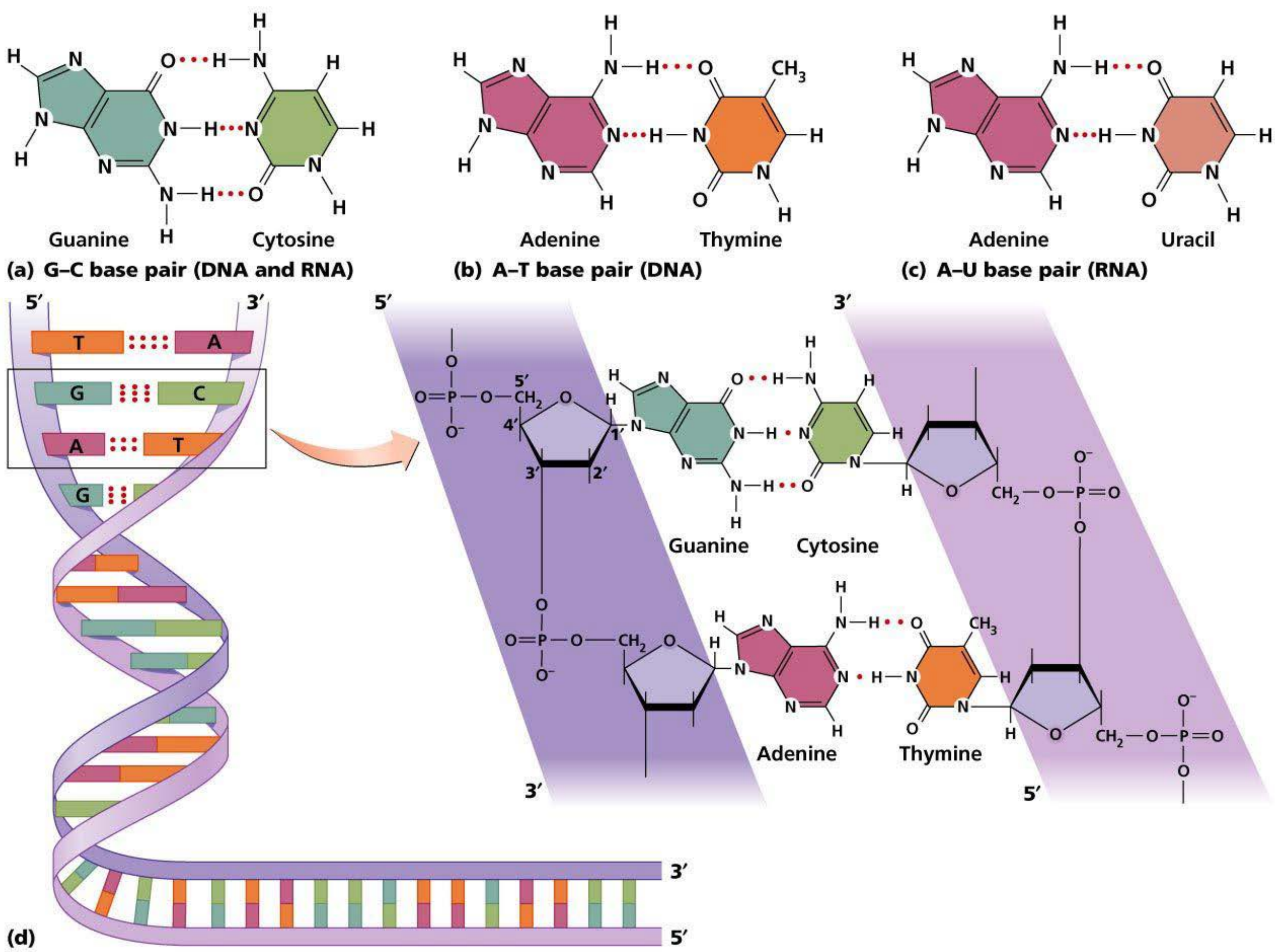
- Hydrogen "bonding" happens with atoms that are electronegative eN-O-F! ☺
- Hydrogen "bonding" is usually depicted with a dotted or dashed line.
- Hydrogen "bonding" is responsible for some of the unique properties of water, including its relatively _________ boiling point.



How do you represent Hydrogen Bonding? Let's look at NH_3.

Hydrogen Bonding: It's in Your DNA!

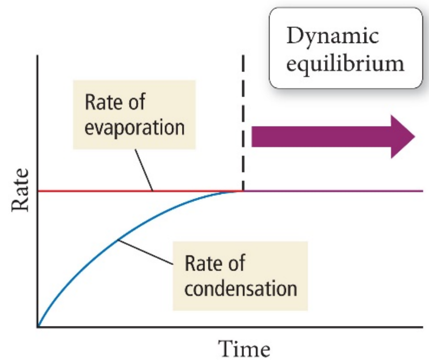
The different number of hydrogen bonds in each complementary base pair (adenine and thymine vs cytosine and guanine) helps ensure that the base pairs will match up correctly!

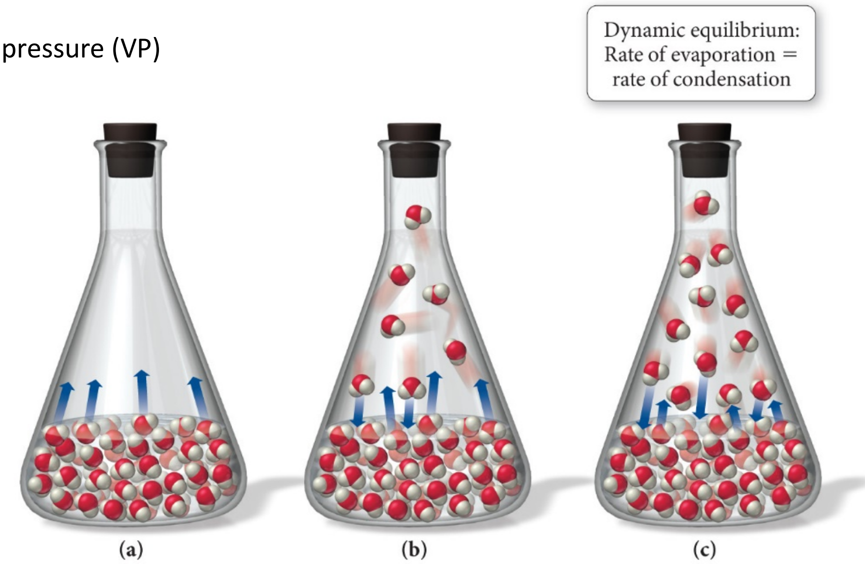


IMFs in Action

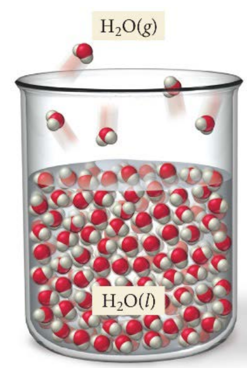
Measure of Intermolecular Forces

1. Vapor Pressure: the pressure exerted by a gas (vapor) when it is in dynamic equilibrium with its liquid (must be in ____________ container!)
    a. The weaker the attractive forces between the molecules, the more molecules will be in vapor (and vice versa).
    b. Thus, ____ IMFs = ____ vapor pressure (VP)





2. Volatility: how quickly a substance evaporates.
    a. The weaker the attractive forces between the molecules, the more quickly and easily molecules will separate from each other and enter the gas phase.
    b. Thus, ____ IMFs = ____ volatility



3. Boiling point: the temperature at which molecules separate from each other in the liquid phase and enter the gas phase.

4. Melting point: temperature at which molecules of a solid have enough thermal energy to overcome IMFs and become a liquid
    a. Thus, ____ IMFs = ____ melting point (MP) = ____ boiling point (BP)

5. Solubility in water: amount of a given substance that will ____________ in water
    a. Strong interactions form between polar/ionic solute particles and polar solvent molecules as they mix → energetically favorable!
    b. Thus, ____ IMFs = ____ solubility in water

Other Properties You Must be able to Explain with IMFs: The **Higher** these are, the **Stronger** the IMFs!

1. Surface tension: energy required to increase the surface area of a liquid
2. Capillary action: spontaneous rising of a liquid in a narrow tube
3. Viscosity: resistance to flow

***Note: Only Vapor Pressure and Volatility have an Inverse Relationship with IMF strength!**

Let's Practice! Draw TWO Lewis dot structures for each of the following molecules. Then, draw dipoles where appropriate and determine the type(s) of intermolecular forces that exist between the molecules.

NH_4^+ Type(s) of IMFs?	CH_3Br Type(s) of IMFs?
H_2O Type(s) of IMFs?	CO_2 Type(s) of IMFs?

More Practice: Draw one EACH of the two compounds given in each example. Then, draw dipoles where appropriate and determine the type(s) of intermolecular forces that exist between the molecules.

NH_4^+ and H_2O	CH_3OCH_3 and H_2O
    Type(s) of IMFs?	 Type(s) of IMFs?

More Practice!

1. What **type** of intermolecular OR intramolecular attractive forces must be overcome for each process below?
   a. $Al(s) \rightarrow Al(l)$

   b. $Al_2O_3(s) \rightarrow 2\ Al(s) + 3/2\ O_2(g)$

2. What is the predominant intermolecular force in CBr_4?
   a. London dispersion forces
   b. Ionic bonding
   c. Dipole-dipole attraction
   d. Hydrogen bonding
   e. Ion-dipole attraction

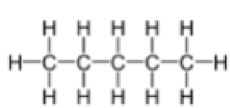
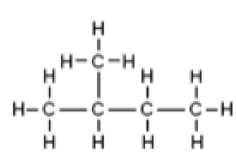
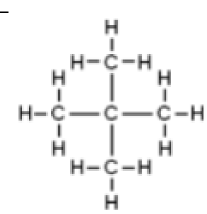
3. Which one of the following derivatives of methane has the highest boiling point?
   a. CBr_4 b. CCl_4 c. CF_4 d. CH_4

4. Based on the molecular mass and dipole moment of the five compounds in the table below, which should have the lowest vapor pressure?

Substance	Molecular Mass (amu)	Dipole Moment (D)
Propane, $CH_3CH_2CH_3$	44	0.1
Dimethylether, CH_3OCH_3	46	1.3
Methylchloride, CH_3Cl	50	1.9
Acetaldehyde, CH_3CHO	44	2.7
Acetonitrile, CH_3CN	41	3.9

   a. $CH_3CH_2CH_3$ b. CH_3OCH_3 c. CH_3Cl d. CH_3CHO e. CH_3CN

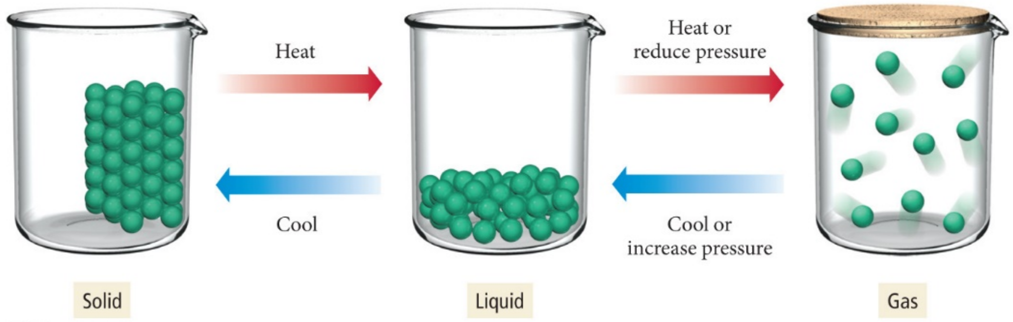
5. Which of the following BEST explains why neopentane has the lowest boiling point?

Common Name	*n*-pentane	isopentane	neopentane
Structure			
Formula	C_5H_{12}	C_5H_{12}	C_5H_{12}
Boiling Point °C	36.0	27.7	9.5

   a. Neopentane is less polarizable due to having fewer electrons.
   b. Neopentane is more polarizable due to having more electrons.
   c. Neopentane has the shortest carbon chains and thus the least surface area.
   d. Neopentane has the shortest carbon chains and thus the most surface area.

IMFs in Action = Heating Curves

KMT: The kinetic-molecular theory is based on the idea that particles of matter are always in motion.



KMT, IMFs and Changes of State: because attractive forces between the molecules are fixed, changing a material's state of matter require changing the amount of kinetic energy the particles have, or limiting their freedom.

1. **Gaseous state**: particles have ______________ freedom of motion.
   a. Their kinetic energy _______________ the attractive forces between the molecules.
2. **Liquid state**: particles have _____________ freedom; they can move around a little within the liquid.
   a. They have enough kinetic energy to overcome __________ of the attractive forces, but not enough to ________________ each other.
3. **Solid state**: particles are locked in place, they _________________ move around.
   a. Although the particles _________________, they do ______ have enough kinetic energy to overcome the attractive forces.

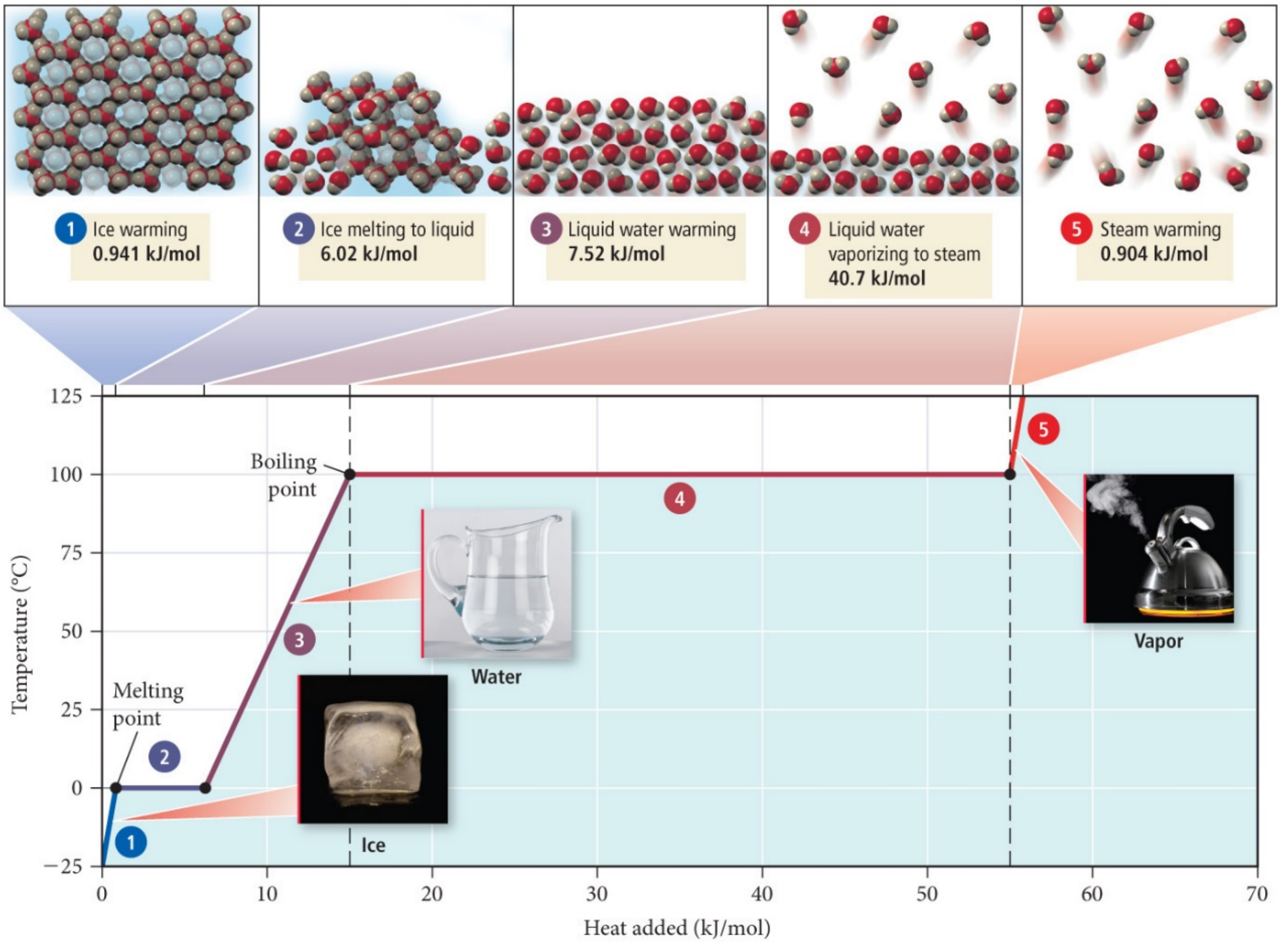
The strength of the attractive forces between particles of a substance determine its state!

- At room temperature, moderate to _____________ attractive forces result in materials that are solids or liquids.
- The __________________ the attractive forces, the _________________ the boiling/ melting point!

State	Density	Shape	Volume	Strength of Intermolecular Forces (Relative to Thermal Energy)
Gas	Low	Indefinite	Indefinite	Weak
Liquid	High	Indefinite	Definite	Moderate
Solid	High	Definite	Definite	Strong

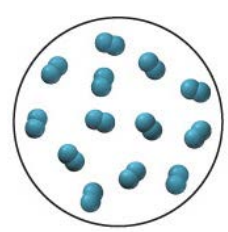
Heating and Cooling Curves

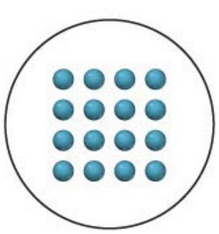
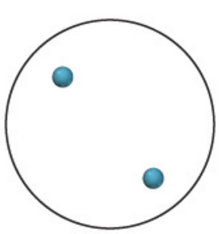
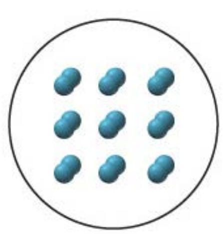
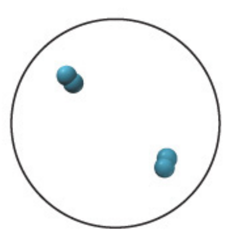
A graph of the temperature of the system versus the amount of heat added.



- In thermochemistry, our focus with heating and cooling curves is on how much ___________ energy is required to change from one state of matter to another.
- For this unit, our focus is on two things:
  - o the relative amount of _______________ energy for each state of matter and
  - o the strength of the ________________ forces (IMFs) holding the particles together in that state

Let's Practice! If liquid nitrogen is shown in the image to the right, which of the images below best depicts nitrogen after it has boiled?



a)  b)  c)  d) 

Solubility and IMFs

Remember solubility rules? 😊 The bolded 3 at the beginning are the only ones you need to memorize, but there are lots and lots of solubility patterns we can observe.

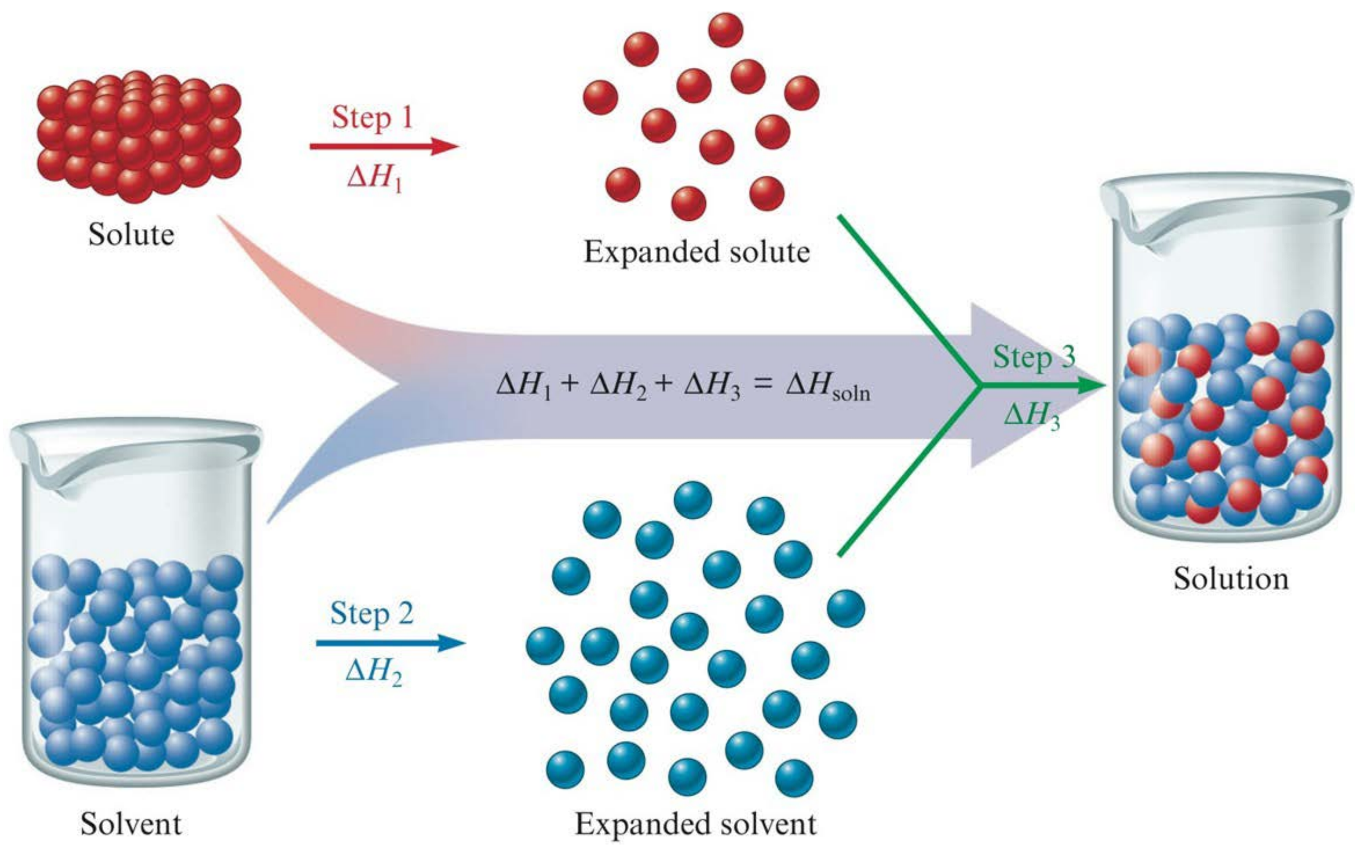
1. **__Always soluble__: _________ metal cations, NH_4^+, NO_3^-** (also ClO_3^-, ClO_4^-, $C_2H_3O_2^-$, HCO_3^-)
2. Generally soluble:
   a. Bromide, chloride, and iodide anions are soluble except when combined with Ag^+, Pb^{2+}, and Hg_2^{2+}.
   b. SO_4^{2-} is soluble except when combined with Sr^{2+}, Ba^{2+}, Pb^{2+}, and Hg_2^{2+}.
3. Generally insoluble:
   a. OH^- and S^{2-} are insoluble except when combined with Ca^{2+}, Sr^{2+}, Ba^{2+}, (and things from rule 1).
   b. CO_3^{2-}, PO_4^{3-}, SO_3^{2-}, and CrO_4^{2-} are insoluble except when combined with things from rule 1.

IMFs help explain these patterns of solubility!

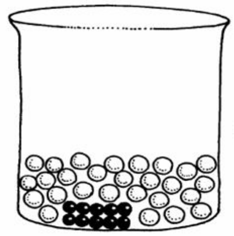
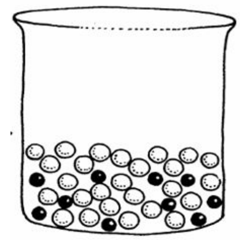
Dissolution depends on the forces of attraction between _______________ and _______________ particles

In order to dissolve a substance, you must:

1. Add energy: Overcome attractions (requires energy = endothermic) *"endo-ing" an attraction is endothermic!*
   a. Solute-solute IMFs (or ion-ion electrostatic attraction, if ionic)
   b. Solvent-solvent IMFs
2. Release energy: Form solute-solvent attractive forces upon mixing (releases energy = exothermic)

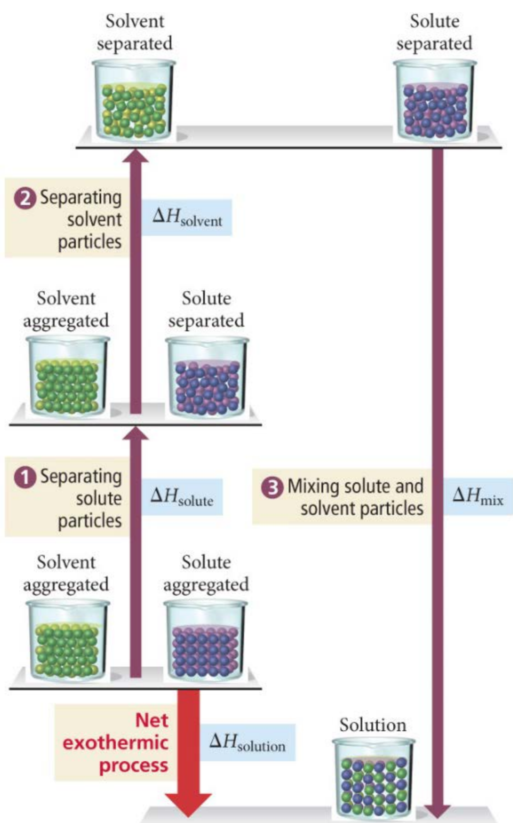
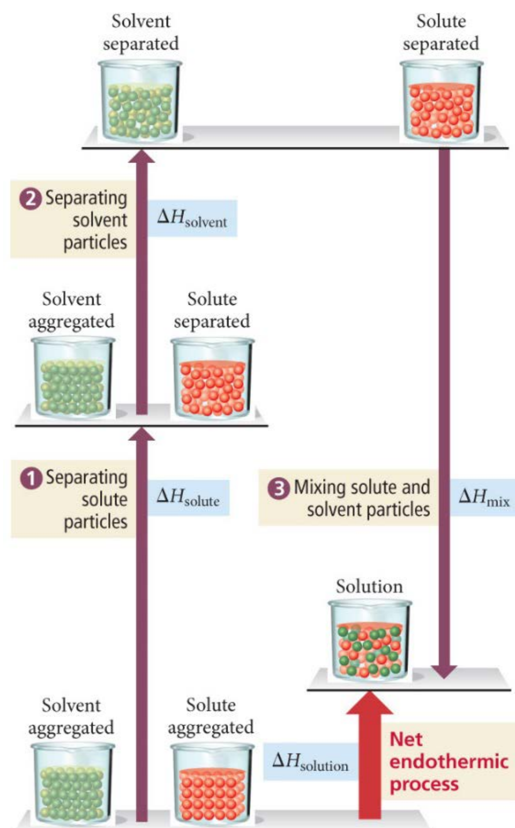


Soluble or Insoluble?

<table>
<tr><th>Soluble</th><th>Insoluble</th></tr>
<tr><td>Higher solute-solvent attractions</td><td>Lower solute-solvent attractions</td></tr>
<tr><td>
Soluble (miscible)</td><td>
Insoluble (immiscible)</td></tr>
</table>

Notice, in general, the solute particles that are **INSOLUBLE** have ____________________ ion charges, which means they have a greater attraction to other solute particles: go, Coulomb's Law, go!

The Thermodynamics of Dissolution

<table>
<tr><th>Exothermic Dissolution ($-\Delta H_{soln}$)</th><th>Endothermic Dissolution ($+\Delta H_{soln}$)</th></tr>
<tr><td>• Heat released when salt dissolve
• Feels warm to the touch</td><td>• Heat absorbed when salt dissolve
• Feels cold to the touch</td></tr>
<tr><td><u>Always</u> thermodynamically favorable ($-\Delta G$) because entropy will always increase
$-\Delta H$, $+\Delta S$</td><td>Thermodynamically favorable at warmer temperatures, depending on increase in entropy
$+\Delta H$, $+\Delta S$</td></tr>
<tr><td>
</td><td>
</td></tr>
</table>

Handy rule of thumb: **"Like dissolves like"**

Great multiple choice trick but does ______ count as explanation on free response!

1. Polar solvents dissolve polar solutes
    - Hydrophilic (polar) groups to watch for: OH, CHO, C=O, COOH, NH_2, and Cl.
2. Non-polar solvents dissolve non-polar solutes
    - Hydrophobic (non-polar) groups to watch for: C – H and C – C.

Many molecules have both hydrophilic and hydrophobic parts; solubility in water becomes a ______________________ between the attraction of the polar groups for water and the attraction of the nonpolar groups for their own kind.

Never use "like dissolves like" to explain a FR on the AP exam: instead, EXPLAIN in terms of structure, IMFs, and energy!

So… how *do* you explain solubility for free response questions?

1. **Identify solute-solvent IMFS**

Type of Substance	Dominant Interaction with Water	Dominant Interaction with a Non-polar Solvent
Ionic	ion-dipole	ion-induced dipole (nope)
Polar + FON	hydrogen bonds	dipole-induced dipole
Polar	dipole-dipole	dipole-induced dipole
Non-polar	dipole-induced dipole	induced dipole-induced dipole (London dispersion forces)

2. **Are solute−solvent attractions ______________ than solute−solute (or solvent−solvent) attractions?**
    - Explain: strong interactions BETWEEN solvent and solute → yes, solute will dissolve!
3. **Solute−solvent attractions ______________ than solute−solute (or solvent−solvent) attractions?**
    - Explain: weak solute−solvent interactions are not as strong as existing solvent−solvent (or solute-solute) attractions, thus solute will _____ dissolve.

Of course, you must be ___________________! Identify **BOTH** solute and solvent by name or formula.

*Note: you do NOT have to explain WHY a given compound can form specific IMFS; it is enough to state them.

Example #1: Can CH_3OH dissolve in water? Why or why not?

Too much "CH_3OH can form hydrogen bonds with water because it has a hydrogen which is covalently bonded to an oxygen, so it will form strong IMFs with water and thus will be able to dissolve in water."

Just right "CH_3OH can form strong hydrogen bonds with water, so it will be able to dissolve."

Not enough "CH_3OH can form strong hydrogen bonds, so it will be able to dissolve in water."

Example #2: Can benzene, C_6H_6, dissolve in water? Why or why not?

Too much "C_6H_6 is non-polar with has a dipole moment of zero, and so it can only form weak dipole-induced dipole interactions with water, which are not as strong as the hydrogen bonds that already exist between water molecules, so C_6H_6 won't dissolve in water."

Just right "C_6H_6 is non-polar and can only form weak intermolecular attractions with water, which are not as strong as the hydrogen bonds that already exist between water molecules, so C_6H_6 won't dissolve in water."

Not enough "C_6H_6 is non-polar, so it won't dissolve in a polar substance like water."

Example FR question: Which is more likely to be soluble in water, liquid methanol (CH_3OH) or liquid hexane (C_6H_{14})? Justify your answer.

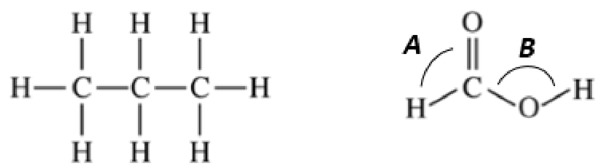
Free Response Practice!

Directions: Use principles of atomic structure, bonding, and intermolecular forces to answer the following questions. Your responses must include specific information about all substances referred to in each part.

1. Ammonia, NH_3, is very soluble in water, whereas phosphine, PH_3, is only moderately soluble in water. Explain.

2. Indicate whether you agree or disagree with the statement in the box below. Justify your answer.

H_2O has a much higher boiling point than H_2S because it has very strong hydrogen bonding forces between its molecules as well as dipole–dipole forces and London dispersion forces, while H_2S has only dipole-dipole and dispersion forces. The stronger hydrogen bonding IMFs between water molecules mean that the bonds between water molecules are harder to break.



Propane

Methanoic Acid

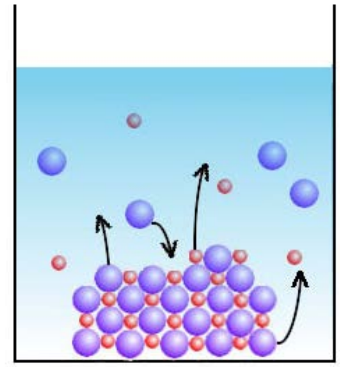
3. The complete structural formulas of propane, C_3H_8, and methanoic acid, HCOOH, are shown above.
   a. In the table below, write the type(s) of intermolecular attractive force(s) that occur in each substance.

Substance	Boiling Point	Intermolecular Attractive Force(s)
Propane	229 K	
Methanoic acid	374 K	

   b. Explain why methanoic acid has a higher boiling point than propane.

   c. Which bond angle would you predict to be larger: angle A (H – C – O) or angle B (C – O – H) in methanoic acid? Justify your answer.

Solubility Equilibria



Up until now, we've pretended that compounds fall into one of two categories: 100% soluble or 100% insoluble. Actually, an equilibrium can exist between a partially soluble substance and its solution.

Solubility Product Constant (_______)

Equilibrium expression for dissolving a solid into ions

Like all K values, K_{sp} is constant at a constant temperature.

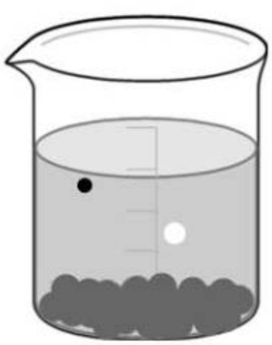
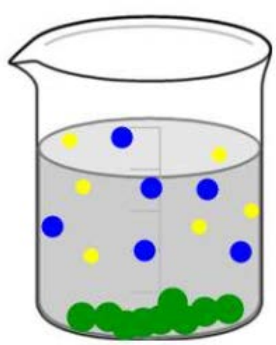
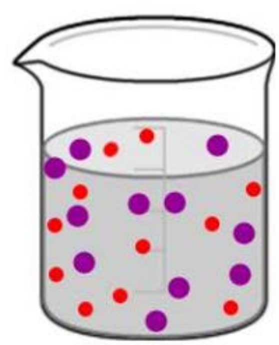
$$\mathrm{AgCl}(s) \rightleftarrows \mathrm{Ag}^{+}(aq) + \mathrm{Cl}^{-}(aq)$$

When the rate at which a solid dissolves into ions is ___________________ to the rate at which ions precipitate back to solid, the system has reached equilibrium.

Let's Practice! Write the *Ksp* expression for each of the following dissolutions:

Salt	Dissociation reaction	K_{sp} Expression
K_2CO_3		
Al_2S_3		

Wait a sec, K_{sp} only has ___________________! But why?

Small K_{sp} (K << 1)	Larger K_{sp}	100% Soluble
• Only a small amount of solid dissociates into ions • Lower solubility	• More solid dissociates into ions • Higher solubility	• All solid dissociates into ions • Dissolves to completion
		

There are three types of 100% soluble ions that you have to memorize:

<u>Always soluble</u>: _________ metal cations, _____, _____

K_{sp} Calculations

ALWAYS write the balanced dissociation reaction and Ksp expression before solving!

Example: When silver sulfide dissolves in pure water, $[S^{2-}] = 3.4 \times 10^{-17}$ mol/L at 25°C. Calculate its K_{sp} value.

Now you try!

1. What is the K_{sp} value of silver phosphate at 25°C, if $[PO_4^{3-}] = 1.8 \times 10^{-18}$ M when dissolved in pure water?

2. What is the concentration of chloride ions that will form in a solution of $PbCl_2$ in pure water? The K_{sp} of lead (II) chloride at 25°C is 1.17×10^{-5}.

But wait! The "____" term, for dissociation, always refers to the amount of ______________ that will dissolve (since the stoichiometric coefficient of the salt will always be _____ in the dissociation reaction. This term has a special name!

<u>Solubility</u> "S" (aka <u>Molar Solubility</u>) = "x" in your K_{sp} RICE Table

How much of a _________ will dissolve per _____ L of solution (Units: M = mol/L)

Solubility is an equilibrium position and therefore ______ change (for example, if you change the number of ions in solution, this will shift the equilibrium position and thus, the solubility).

- Larger molar solubility values suggest _________ dissociation into ions and <u>greater</u> solubility.
- Smaller molar solubility values suggest _________ dissociation into ions and <u>lower</u> solubility.
- **<u>Note</u>: when comparing solubility of given compounds, you should <u>compare their molar solubility values</u> and NOT their K_{sp} values!**

Now we can answer problems using the same math we used on the previous page. The terminology will be different, but the calculations are the same! Let's try some:

3. The molar solubility of barium fluoride is at 25°C is 2.45 x 10^{-5} M. Calculate K_{sp}.

4. Calculate the molar solubility of nickel (II) carbonate, which has a K_{sp} of 1.4 × 10^{-7} at 25°C.

How Much Will Dissolve?

You can use molar solubility to determine how many ____________ of a solid will dissolve in a quantity of water!

5. How many grams of iron (II) hydroxide can dissolve in 500. mL of water? Its $K_{sp} = 4.87 \times 10^{-17}$ at 25°C.

6. The molar solubility of $PbCl_2$ in pure water is 1.43×10^{-2} M at 25ºC.
    a. Write the equilibrium constant expression for the dissolving of $PbCl_2(s)$.

    b. How many grams of $PbCl_2$ can dissolve into 200. mL of pure water?

    c. What change could be made to increase amount of $PbCl_2$ that will dissolve?

Multiple Choice Practice!

7. The solubility product, K_{sp}, of AgCl is 1.8×10^{-10}. Which of the following expressions is equal to the molar solubility of AgCl?

   a. $(1.8 \times 10^{-10})^2$ molar

   b. $\frac{1.8\times10^{-10}}{2}$ molar

   c. 1.8×10^{-10} molar

   d. $\sqrt{1.8 \times 10^{-10}}$ molar

8. Which of the following is equal to the solubility product, K_{sp}, of Ag_2CO_3?

   a. $K_{sp} = [Ag^+][CO_3^{2-}]$

   b. $K_{sp} = [Ag^+][CO_3^{2-}]^2$

   c. $K_{sp} = [Ag^+]^2[CO_3^{2-}]$

   d. $K_{sp} = [Ag^+]^2[CO_3^{2-}]^2$

9. If the solubility of BaF_2 is equal to x, which of the following expressions is equal to the solubility product, K_{sp}, of BaF_2?

   a. x^2

   b. $2x^2$

   c. $2x^3$

   d. $4x^3$

10. In a saturated solution of Na_3PO_4, $[Na^+]$ = 0.30 M. What is the molar solubility of Na_3PO_4?

    a. 0.10 M

    b. 0.30 M

    c. 0.60 M

    d. 0.90 M

11. What is the maximum mass of AgCl can dissolve in 100. mL of pure water at 25°C? The molar solubility of AgCl is 1.3×10^{-5}.

    a. 1.9×10^{-4} g

    b. 1.9×10^{-5} g

    c. 9.1×10^{-9} g

    d. 9.1×10^{-6} g

The Common Ion Effect

Remember: solubility can change if you change reaction conditions!

- Le Châtelier's principle predicts that a salt will become __________ soluble in a solution that already contains one of its own ions already dissolved: what's known as a _______________ ion.
- The presence of a common ion acts like increasing the concentration of a _________________ ion in the salt dissolution, causing the system to shift ___________ to establish equilibrium (towards the ________ side).

Example:

1. Circle any of the following compounds that contain a common ion to $MgCl_2$:

   AgF $NaCl$ $Mg(OH)_2$ $AlCl_3$ Al_2S_3

2. Which of the compounds above, if present in the solution in equal concentration, would reduce the solubility of $MgCl_2$:
   a. the most? Why?
   b. the least? Why?

The Effect of pH on Solubility

The common ion effect predicts that when a salt contains ions that can act as an acid or a base, the solubility of that salt will be affected by changes in ________.

Example: $Fe(OH)_3 \rightleftharpoons Fe^{3+}(aq) + 3\ OH^{-}(aq)$

3. Will iron (III) hydroxide be more, less, or equally soluble in a **basic** solution (when compared to its solubility in pure water)? Explain.

4. Will iron (III) hydroxide be more, less, or equally soluble in an **acidic** solution (when compared to its solubility in pure water)? Explain.

Common Ion Calculations

5. The K_{sp} of $NiCO_3$ is 1.4 × 10−7 at 25°C.
   a. Calculate the molar solubility of nickel (II) carbonate in pure water.
   b. Calculate the molar solubility of nickel (II) carbonate in solution containing 0.100 M Na_2CO_3.

6. Calculate the molar solubility of SrF_2 in a solution containing 0.400 M NaF. The K_{sp} of SrF_2 is 7.9×10^{-10} at 25°C.

Common Ion Applications: Achieving a Desired Concentration

You can use the common ion effect to control the ____________________ of the **non-common** ion in solution! Let's see how this chemical magic works.

7. In pure water at 25°C, the K_{sp} of $CaCO_3$ is 3.8×10^{-9}. $Ca(NO_3)_2$ is added to 1.00 L of a saturated solution of $CaCO_3$ at 25°C until the $[CO_3^{2-}]$ is reduced to 2.3×10^{-7} M. How many moles of $Ca(NO_3)_2$ are dissolved in solution at the point when $[CO_3^{2-}] = 2.3 \times 10^{-7}$ M? (Assume the added $Ca(NO_3)_2$ has a negligible effect on the total volume of solution.)

8. Given a 2.00 L saturated solution of $Cu(IO_3)_2$, $K_{sp} = 1.4 \times 10^{-7}$, how many moles of $NaIO_3$ would need to be dissolved in solution to reduce $[Cu^{2+}]$ to 6.0×10^{-5} M? (Assume the added $NaIO_3$ does not appreciably change the total volume of solution.)

Even more practice!

9. Copper(I) bromide has a measured solubility of 2.0×10^{-4} mol/L at 25°C. Calculate its K_{sp} value.

10. The K_{sp} value for copper(II) iodate, $Cu(IO_3)_2$, is 1.4×10^{-7} at 25°C. What is the maximum mass, in grams, of copper (II) iodate that can dissolve in 500. mL of water?

11. In pure water at 25°C, the molar solubility of $PbCl_2$ is 1.3×10^{-6} and the K_{sp} is 1.6×10^{-5}. LiCl is added to 5.00 L of a saturated solution of $PbCl_2$ at 25°C until the $[Pb^{2+}]$ is reduced to 4.5×10^{-4} M. How many moles of chloride ions are dissolved in solution at the point when $[Pb^{2+}] = 4.5 \times 10^{-4}$ M? (Assume the added LiCl has a negligible effect on the total volume of solution.)

Multiple Choice Practice!

Use the following information to answer questions 12−14.

150 mL of saturated SrF_2 solution is present in a 250 mL beaker at room temperature. The molar solubility of SrF_2 at 298 K is 1.0 x 10^{-3} M.

12. What are the concentrations of Sr^{2+} and F^- in the beaker?
    a. $[Sr^{2+}] = 1.0 \times 10^{-3}$ M; $[F^-] = 1.0 \times 10^{-3}$ M
    b. $[Sr^{2+}] = 1.0 \times 10^{-3}$ M; $[F^-] = 2.0 \times 10^{-3}$ M
    c. $[Sr^{2+}] = 2.0 \times 10^{-3}$ M; $[F^-] = 1.0 \times 10^{-3}$ M
    d. $[Sr^{2+}] = 2.0 \times 10^{-3}$ M; $[F^-] = 2.0 \times 10^{-3}$ M

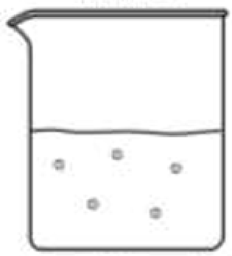
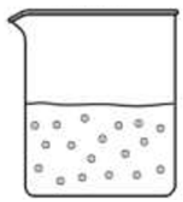
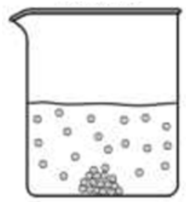
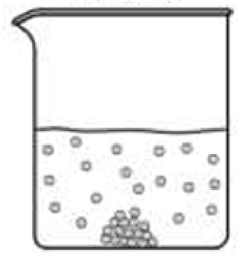
13. What would be the effect on $[Sr^{2+}]$ if some NaF(s) was added to the beaker?
    a. $[Sr^{2+}]$ would remain unchanged; neither ion in NaF(s) is common to Sr^{2+}
    b. $[Sr^{2+}]$ would increase; more Sr^{2+} ions would be needed to balance the additional F^- ions to re-establish equilibrium.
    c. $[Sr^{2+}]$ would decrease; the additional fluoride ions would cause the system to shift left to re-establish equilibrium.
    d. $[Sr^{2+}]$ would decrease; the additional Na^+ would cause an excess of positive charge, and the system would shift left to reduce overall positive charge.

14. Calculate the solubility product for SrF_2 at 25°C.
    a. 2 x 10^{-9}
    b. 2 x 10^{-6}
    c. 4 x 10^{-9}
    d. 4 x 10^{-6}

Will a Precipitate Form? A Task for K vs Q!

Precipitation occurs when the concentrations of ions is ________________ than the solubility of the ionic compound.

Compare the value of Q with given K_{sp} to determine if a precipitate will form!

K > Q	**K = Q**	**K < Q**
Unsaturated Solution	Saturated Solution	Saturated Solution with extra
System will shift right to reach equilibrium $\mathrm{AgCl(s)} \rightleftarrows \mathrm{Ag^+}(aq) + \mathrm{Cl^-}(aq)$	At equilibrium $\mathrm{AgCl(s)} \rightleftarrows \mathrm{Ag^+}(aq) + \mathrm{Cl^-}(aq)$	System will shift left to reach equilibrium $\mathrm{AgCl(s)} \leftrightarrows \mathrm{Ag^+}(aq) + \mathrm{Cl^-}(aq)$
More solid will dissolve until K = Q.	Solid will both dissolve and precipitate at the same rate.	**More solid will precipitate** until K = Q.
 no precipitate	  maybe precipitate	 yes precipitate

Important Ideas to Note:

1. If ______ solid is present, the solution is at equilibrium (a ____________________ solution)
2. Ion concentration, [ions], is **independent** of volume when at equilibrium (for instance, in a _________________ solution).
3. If ions are present that could form _______________ salts, the solid with the ______________ molar solubility will form.

Solubility Equilibrium Translation Guide

1. <u>Solubility product constant</u> = K_{sp} (aka the equilibrium constant for solubility)
2. <u>Molar solubility</u> = x from RICE table (aka how many moles of a solid will dissolve in 1.0 L, units = M = mol/L)
3. <u>Saturated</u> = equilibrium (aka a solution has dissolved as many ions as can fit, any extra will precipitate)

Let's Practice!

1. A chemist makes a 2.0 L saturated solution of $Ba_3(PO_4)_2$ solution, which has a K_{sp} = 6.0 × 10^{-39}.
   a. What is the concentration of Ba^{2+} ions in solution?

b. After two days of sitting on the counter, some liquid has evaporated from the solution. Did $[Ba^{2+}]$ increase, decrease, or remain the same? Justify your answer.

c. The chemist adds 3.00 g of solid $(NH_4)_3PO_4$ to the original saturated solution of $Ba_3(PO_4)_2$. Did $[Ba^{2+}]$ increase, decrease, or remain the same? Justify your answer.

2. A solution containing lead (II) nitrate is mixed with one containing sodium bromide to form a solution that is 0.0150 M in $Pb(NO_3)_3$ and 0.00350 M NaBr. Does a precipitate form in this newly mixed solution? (K_{sp} of $PbBr_2$ $= 4.67 \times 10^{-6}$)

3. The K_{sp} value for lead (II) bromide, $PbBr_2$, is 4.6×10^{-6} at 25°C. What is the maximum mass, in grams, of $PbBr_2$ that can dissolve in 1.50 L of water?

4. A student mixes 15.0 mL of 0.015 M sodium iodide solution, NaI, with 5.00 mL of 0.0025 M $Pb(NO_3)_2$. The K_{sp} of PbI_2 is 8.5 x 10^{-9} M. What will the student observe? Justify your answer with calculations.

5. Sodium carbonate is added to a 0.0024 M solution of the nickel (II) ion. If $[Na_2CO_3]$ = 1.0 x 10^{-4} M, will a precipitate form? (The K_{sp} of nickel (II) carbonate is 6.6 x 10^{-9}.)

6. Calculate the molar solubility of $Ba_3(PO_4)_2$, which has a $K_{sp} = 6.0 \times 10^{-39}$.

Multiple Choice Practice!

7. 150 mL of saturated SrF_2 solution is present in a 250 mL beaker at room temperature. If some of the solution evaporates overnight, which of the following will occur?

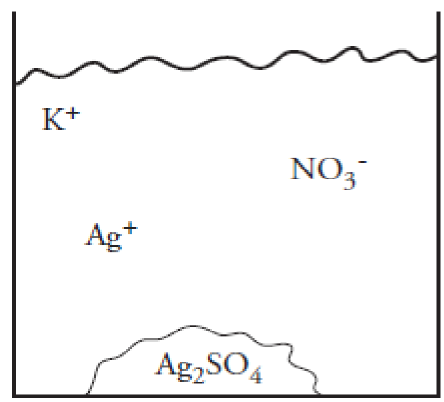
   a. The mass of the solid and the concentration of the ions will remain the same.
   b. The mass of the solid and the concentration of the ions will increase.
   c. The mass of the solid will decrease, and the concentration of the ions will remain the same.
   d. The mass of the solid will increase, and the concentration of the ions will remain the same.

8. A student added 1 liter of a 1.0 M KCl solution to 1 liter of a 1.0 M $Pb(NO_3)_3$ solution. A lead chloride precipitate formed, and nearly all of the lead ions disappeared from solution. Which of the following lists the ions remaining in the solution in order of decreasing concentration?

   a. $[NO_3^-] > [K^+] > [Pb^{2+}]$
   b. $[NO_3^-] > [Pb^{2+}] > [K^+]$
   c. $[K^+] > [Pb^{2+}] > [NO_3^-]$
   d. $[K^+] > [NO_3^-] > [Pb^{2+}]$

Use the following information to answer questions 8−10.

Silver sulfate, Ag_2SO_4, has a solubility product constant of 1.0×10^{-5}. The diagram to the right shows the products of a precipitation reaction in which some silver sulfate was formed.



9. What is the identity of the excess reactant?

   a. $AgNO_3$
   b. Ag_2SO_4
   c. $NaNO_3$
   d. Na_2SO_4

10. If the beaker above was left uncovered for several hours:

    I. Some of the Ag_2SO_4 would dissolve.
    II. Additional Ag_2SO_4 would precipitate.
    III. $[Ag^+]$ would remain constant.

    a. I only
    b. II only
    c. II and III
    d. I and III

11. Which ion concentration below would have led the precipitate to form?

    a. $[Ag^+] = 0.01$ M, $[SO_4^{2-}] = 0.01$ M
    b. $[Ag^+] = 0.10$ M, $[SO_4^{2-}] = 0.01$ M
    c. $[Ag^+] = 0.01$ M, $[SO_4^{2-}] = 0.10$ M
    d. It is impossible to determine without knowing the total volume of the solution.

Solubility Summary Sheet

Solubility Language	Normal Equilibrium Language
solubility product constant (K_{sp})	equilibrium constant (K)
molar solubility	x (from RICE table)
saturated solution	system at equilibrium

When you need to solve for molar solubility

Given?	Asked to find?	Strategy
K_{sp}	molar solubility	1. Write K_{sp} expression using ions produced when solid dissolves. 2. Substitute x values from mini-RICE table 3. Solve for x
1. K_{sp} OR 2. Concentration of all ions in a saturated solution	# of grams that can dissolve	1. Write K_{sp} expression using ions produced when solid dissolves. 2. Substitute x values from mini-RICE table 3. Solve for x (in M = mol / L) 4. Use the volume of solution to calculate moles that can dissolve 5. Use molar mass to convert to grams

When you don't need to solve for molar solubility

Given?	Asked to find?	Use:
Concentration of all ions in a saturated solution	K_{sp}	1. Write K_{sp} expression using ions produced when solid dissolves. 2. Substitute given concentrations and solve for K_{sp}.
other ion concentrations at equilibrium <u>and</u> K_{sp}	Concentration of ONE ion in a saturated solution	1. Write K_{sp} expression using ions produced when solid dissolves. 2. Plug in known values, solve for unknown concentration.
ion concentrations when solutions are added or mixed <u>and</u> K_{sp}	If a precipitate will form	1. Write Q_{sp} expression using ions produced when solid dissolves. 2. Substitute given concentrations and solve for Q_{sp}. 3. Compare K and Q: • K > Q = no precipitate • K < Q = yes precipitate